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ROBUST COMMUNICATION FOR VEHICLE PLATOONING USING ACO AND SPIDER WEB TOPOLOGY

A MAJOR PROJECT REPORT submitted in partial fulfilment of the requirements for the
award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND TELECOMMUNICATION ENGINEERING

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DECLARATION

We, the students of Electronics and Telecommunication Engineering, Barak Valley Engineering College, declare that the work entitled "**ROBUST COMMUNICATION FOR VEHICLE PLATOONING USING ACO AND SPIDER WEB TOPOLOGY**" has been successfully completed under the guidance of **KAMAL HASAN LASKAR, Assistant Professor, Department of Electronics and Telecommunication Engineering, Barak Valley Engineering College.**

We declare that the report has been prepared without resorting to plagiarism. We have adhered to the principles of academic honesty and integrity. No falsified or fabricated data have been presented in the report. Further, the matter embodied in this project report has not been submitted previously by anybody for the award of any degree or diploma to any university.

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ABSTRACT

Vehicle platooning is an important application of intelligent transportation systems in which a group of vehicles moves in a coordinated formation. In such systems, reliable Vehicle-to-Vehicle communication is essential because the master vehicle must transmit trajectory, speed, position, and control-related information to the slave vehicles. If communication fails, the platoon may experience unsafe spacing, delayed response, or loss of coordination. Traditional star-based communication is simple but vulnerable because each slave vehicle depends directly on the master vehicle.

The proposed system investigates robust communication for vehicle platooning using **Ant Colony Optimization** and a **Spider-Web Network Topology**. The spider-web topology provides redundant multi-hop communication paths, while ACO selects routes adaptively based on pheromone strength, reliability, congestion, and latency. The project repository implements the system in Python using a graph model with one master vehicle and six slave vehicles. Each communication edge is assigned latency, pheromone, reliability, and congestion attributes. Dynamic link failures and changing network conditions are simulated, and ACO routing is compared with a baseline shortest-path method.

The representative simulation results show **PDR (ACO) = 0.975**, **PDR (Baseline) = 0.645**, **ACO average latency = 13.879 ms**, and **Baseline average latency = 10.477 ms**. The results indicate that ACO improves packet delivery reliability by using alternate routes when static paths degrade. However, the improved reliability may increase latency and redundancy because the selected path may contain additional hops. The proposed system therefore demonstrates an important engineering trade-off between reliability, latency, and communication overhead.

Keywords: Vehicle platooning, V2V communication, Ant Colony Optimization, spider-web topology, packet delivery ratio, latency, redundancy, adaptive routing.

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CHAPTER 1: INTRODUCTION

1.1 Background

Vehicle platooning is a cooperative driving technique in which multiple vehicles travel together as a coordinated group. A platoon typically contains a leader or master vehicle and several follower or slave vehicles. The leader defines the desired trajectory, velocity profile, and manoeuvre information, while the following vehicles maintain safe inter-vehicle distance and coordinated movement. The concept is significant in modern intelligent transportation systems because it can improve road capacity, reduce aerodynamic drag, reduce fuel consumption, and support smoother traffic flow.

In a platooning system, each vehicle is not an isolated unit. The safety and efficiency of the entire group depend on continuous information exchange. A follower vehicle must know when the leader accelerates, decelerates, changes lane, or detects an obstacle. This communication must be timely and reliable because even a small delay can affect the spacing between vehicles. Unlike ordinary vehicular communication, platooning communication directly influences vehicle control decisions. Therefore, communication must be treated as a safety-critical component of the system.

The project titled **"Robust Communication for Vehicle Platooning Using Ant Colony Optimization and Spider-Web Network Topology"** focuses on the communication layer of a platoon. It does not attempt to implement full autonomous driving control. Instead, it studies how messages can be routed from the master vehicle to multiple slave vehicles when network conditions are dynamic. The project is inspired by the IEEE INDISCON 2025 paper on robust communication for next-generation vehicle platooning and implements the core idea through a Python simulation.

Vehicle-to-Vehicle communication allows vehicles to exchange state and environment information without depending entirely on roadside infrastructure. In platooning, V2V messages may include position, speed, acceleration, heading angle, braking commands, object detection alerts, local path updates, and emergency warnings. These messages allow vehicles to synchronise actions and maintain formation stability.

The communication problem becomes more complex when the platoon size increases or when the environment contains interference, mobility, congestion, and temporary link failures. A direct link between the master and a slave vehicle may not always be available. If the system uses only a star topology, failure of a direct link can isolate a vehicle even though other neighbouring vehicles may still be able to relay the message. This motivates the use of

multi-hop communication, where a packet can travel through intermediate vehicles.

Vehicular communication networks are dynamic because vehicles move, distances change, and wireless link quality fluctuates. A communication link that is reliable at one moment may become weak due to distance, obstruction, interference, or congestion. The following challenges are especially relevant to platooning:

- **Link instability:** Wireless links may fail temporarily due to mobility or signal degradation. - **Latency sensitivity:** Platooning messages must be delivered within a bounded time interval. - **Packet loss:** Lost control messages can reduce coordination quality. - **Congestion:** Multiple vehicles transmitting simultaneously may increase delay. - **Scalability:** Communication should remain reliable as the number of vehicles increases. - **Single-point dependence:** Star topology depends heavily on direct communication from the master.

These challenges require routing mechanisms that can adapt to changing network conditions rather than relying on a static path.

1.2 Literature Survey

The literature review studies three connected areas: vehicle platooning communication, Ant Colony Optimization in routing, and spider-web-inspired network design. The objective is to understand why the project combines ACO with a redundant multi-hop topology. The IEEE INDISCON 2025 paper is used as the central reference because the repository project is based on that paper.

1.2.1 Comparison of Related Works

Table 2.1: Summary of related work.

Work	Focus Area	Relevance to This Project
Dorigo et al. [2]	Ant Colony Optimization foundations	Provides the bio-inspired routing principle used in <code>aco.py</code> .
Wei et al. [3]	Virtual spider-web routing model	Supports the use of spider-web-inspired redundant topology.
Hou et al. [4]	Large-scale vehicle platooning	Provides broader context on platoon scheduling and planning challenges.
Jabbar and Pinandito [5]	ACO vs Dijkstra routing	Supports comparison between adaptive and shortest-path routing.
Bharadwaaj et al. [1]	ACO-Spider-Web platooning communication	Primary research basis for this project.
Python NetworkX [6]	Graph modelling	Supports the graph representation used by the repository.

Work	Focus Area	Relevance to This Project
Matplotlib [7]	Visualization	Supports graph generation and result plots.

1.2.2 Research Gap Identification

The research gap addressed by this project is the need for an adaptive and fault-tolerant communication layer in vehicle platooning. Star topology is easy to implement but is weak under failures. Static shortest-path routing is efficient under stable conditions but may not adapt well when the selected path degrades. ACO combined with spider-web topology addresses this gap by maintaining multiple candidate paths and continuously reinforcing successful routes.

The project also reveals a practical engineering gap: improving reliability can increase latency and redundancy. Therefore, robust communication design must consider multiple metrics rather than optimizing only for shortest path length.

1.3 Motivation

Adaptive routing is motivated by the observation that the best communication path changes over time. In a static routing method, the shortest path is computed once and reused. This method can be efficient under stable conditions, but it is vulnerable when one edge in the path fails. A dynamic method can evaluate multiple candidate routes and select the most suitable one based on current conditions.

Ant Colony Optimization is suitable for this project because it is a distributed and bio-inspired search method. Natural ants discover efficient routes between a nest and food source through pheromone trails. Artificial ACO algorithms use a similar idea: successful paths receive stronger pheromone values, while weaker or unused paths lose attractiveness through evaporation. In the context of vehicle platooning, pheromone values represent learned communication utility. When a path delivers packets successfully with reasonable latency, it becomes more attractive for future transmissions.

The repository implements a graph-based platooning communication simulation. The master vehicle is represented by node **M**, and six slave vehicles are represented by nodes **S1** to **S6**. The graph uses a spider-web-inspired structure that contains both direct master-to-slave links and slave-to-slave relay links. Each edge contains the following attributes:

- **weight**: communication latency or path cost;
- **pheromone**: learned utility used by ACO;
- **reliability**: probability-related quality indicator;
- **congestion**: multiplier that increases effective cost under network load.

The simulation repeatedly sends packets from the master to all slave vehicles. During each run, random link failures, latency variation, congestion changes, and reliability fluctuation are introduced. The ACO algorithm selects routes dynamically, while the baseline method uses precomputed shortest paths. The system reports Packet Delivery Ratio, average latency, and average redundancy.

ACO-Spider-Web Vehicle Platooning Architecture

The project models resilient V2V communication between one master vehicle and six slave vehicles.

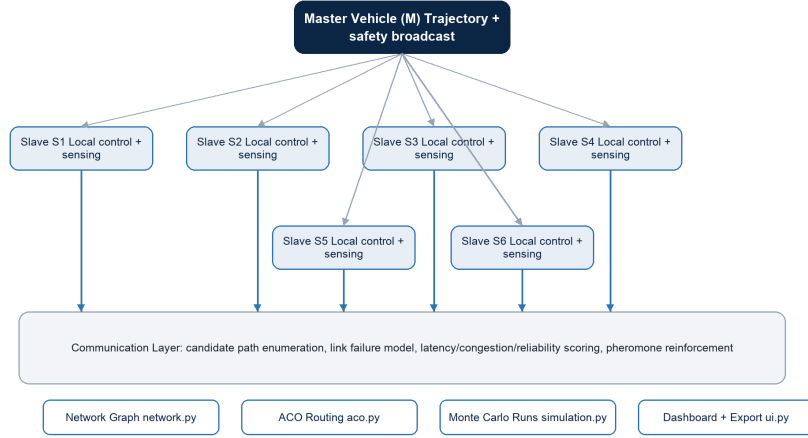


Fig. 1.1: Conceptual architecture of the ACO-Spider-Web platooning communication system.

The scope of this project is limited to communication simulation and routing analysis. It does not implement physical vehicle control, real wireless communication, or full sensor fusion. However, it provides a technically meaningful abstraction for studying message delivery under dynamic conditions. The significance of the project lies in demonstrating that redundant topology and adaptive routing can improve reliability in platooning communication.

The project is academically significant because it combines concepts from intelligent transportation systems, graph theory, bio-inspired optimization, and simulation-based performance analysis. It is also practically relevant because future autonomous platoons will require communication systems that are resilient to failures and congestion.

1.4 Problem Formulation

1.4.1 Problem Statement

The problem addressed in this project is the design and evaluation of a robust communication mechanism for vehicle platooning in which a master vehicle must deliver coordination messages to multiple slave vehicles under dynamic network conditions. The network may experience link failures, fluctuating latency, congestion, and changing reliability. A static or centralized communication method may fail to deliver packets when direct links or

precomputed paths become unavailable.

The project therefore asks the following research question:

How can adaptive routing using Ant Colony Optimization and a spider-web network topology improve packet delivery reliability in vehicle platooning while managing latency and redundancy trade-offs?

1.4.2 Objectives

The objectives of the project are listed in Table 3.1.

Table 3.1: Project objectives and expected outputs.

Objective No.	Objective	Expected Output
O1	Model a vehicle platoon as a communication graph	Graph with master node M and slave nodes S1 - S6
O2	Implement a spider-web-inspired topology	Redundant multi-hop links between vehicles
O3	Implement ACO-based route selection	Probabilistic path selection using pheromone and heuristic quality
O4	Simulate dynamic network conditions	Link failures, congestion variation, latency variation, and reliability fluctuation
O5	Compare ACO with a baseline method	PDR, latency, and redundancy comparison
O6	Generate visual evidence	Plots, UI screenshot, animation screenshot, and report figures
O7	Prepare academic documentation	Final-year report with equations, figures, tables, and references

1.4.3 Expected Contributions

The expected contributions of the project are:

1. A Python-based simulation model for robust platooning communication. 2. A spider-web topology implementation with redundant relay paths. 3. An ACO routing algorithm that considers latency, congestion, reliability, and pheromone history. 4. A comparison between adaptive routing and a static baseline shortest-path method. 5. A dashboard interface for running simulations and exporting results. 6. Static and animated visualizations of routing behaviour. 7. A research-style discussion of reliability-latency trade-offs.

1.4.4 Assumptions

The simulation uses the following assumptions:

- The platoon contains one master vehicle and six slave vehicles. - The communication network can be represented as an undirected graph. - Edge weights represent latency or communication cost. - Reliability is represented as a numeric edge attribute between 0.5 and 1.0 after clamping. - Congestion acts as a multiplicative factor on effective link cost. - Packet delivery is considered successful if a valid path exists and is selected. - The baseline method uses precomputed shortest paths.

1.4.5 Limitations

The project is a simulation prototype and has the following limitations:

- It does not implement real wireless communication protocols. - It does not integrate with CARLA or SUMO in the current repository. - It does not use physical vehicle dynamics or control-loop deadlines. - The graph size is small and fixed. - Random simulation parameters are not controlled by a seed in the main code. - Tests are smoke-style scripts rather than formal assertion-based unit tests.

These limitations are not defects in the project concept. They identify the boundary between a final-year simulation prototype and a full vehicular communication research platform.

CHAPTER 2: THEORY

2.1 Vehicle Platooning Communication Systems

Vehicle platooning requires more than ordinary data exchange. Communication directly supports motion coordination, which means reliability and latency must be evaluated together. A platoon may be stable only when the follower vehicles receive timely information about the master vehicle and neighbouring vehicles. V2V communication is therefore used to exchange dynamic state information.

Traditional platooning systems often assume a leader-follower architecture. The leader broadcasts state information, and followers react accordingly. This assumption is simple, but it may become fragile if the communication topology is centralized. When all vehicles depend only on the leader, a failed leader-to-follower link can affect the entire formation. Multi-hop communication can reduce this risk by allowing messages to be relayed through neighbouring vehicles.

2.2 Ant Colony Optimization in Networking

Ant Colony Optimization was introduced by Dorigo and later developed into a general metaheuristic for combinatorial optimization. In routing applications, ACO is attractive because it naturally balances exploration and exploitation. Exploration allows the algorithm to discover alternative routes, while exploitation allows it to reuse historically successful routes.

In communication networks, each edge can be assigned a pheromone value. Routes with lower delay, fewer failures, or higher delivery success receive more reinforcement. Pheromone evaporation prevents the algorithm from permanently selecting a route that was good in the past but has become poor due to changing network conditions. This makes ACO suitable for dynamic networks such as VANETs and platooning networks.

2.3 Spider-Web Network Topology

The spider-web topology is inspired by the redundant structure of a natural web. Instead of connecting every slave node only to the master, nodes are connected through multiple neighbouring paths. This design supports fallback communication. If one link fails, the network can still use an alternate path. In the project repository, the graph includes direct links from M to S1, S2, and S3, and additional slave-to-slave links such as S1-S2, S2-S3, S4-S5, S5-S6, S2-S6, and S3-S5.

The spider-web structure is especially useful when combined with ACO. ACO requires multiple candidate paths in order to choose adaptively. A star topology provides fewer

alternatives, while the spider-web topology provides enough path diversity for the optimization mechanism to be meaningful.

2.4 IEEE Reference Paper Foundation

The IEEE INDISCON 2025 paper "**On Robust Communication for Next-Gen Vehicle Platooning**" proposes a hybrid communication framework combining a spider-web-inspired network topology with ACO-based routing. The paper identifies a gap in platooning research: many studies focus on vehicle control, perception, and object detection, but assume that the communication layer is reliable. The paper argues that this assumption is not sufficient because real vehicular environments include link failures, interference, latency variation, and packet loss.

The paper evaluates the proposed communication strategy using metrics such as Packet Delivery Ratio, latency, failure rate, offline rate, and redundancy. It reports that ACO-Spider-Web communication improves success rate and reduces latency compared with a star topology. This project implements a related model in Python and evaluates it through a smaller graph simulation.

CHAPTER 3: METHODOLOGY

3.1 Proposed Methodology Overview

The proposed methodology combines a spider-web network topology with Ant Colony Optimization. The topology provides multiple communication paths, while ACO selects paths adaptively based on learned and current link quality. The methodology is implemented in the repository through `network.py`, `aco.py`, and `simulation.py`.

Communication Topology: Star Baseline vs. Spider-Web Redundancy

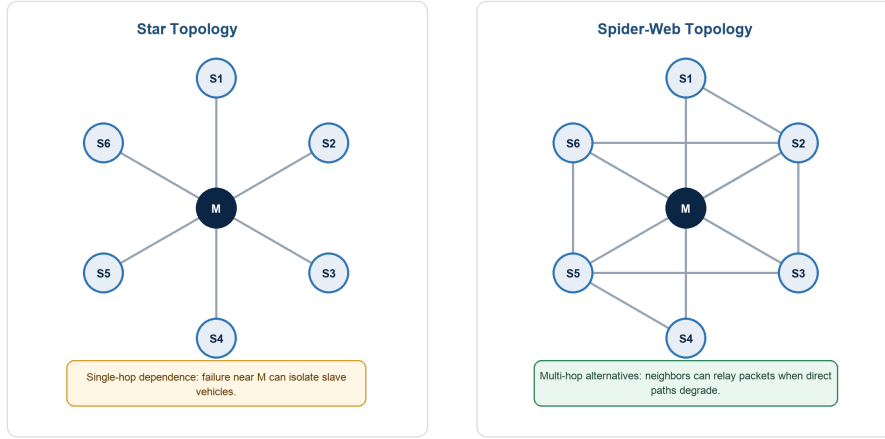


Fig. 3.1: Star topology and spider-web topology comparison.

3.2 Spider-Web Topology Design

The spider-web topology contains direct and indirect communication links. The master vehicle is connected to selected nearby slave vehicles, while slave vehicles are also connected to neighbouring slave vehicles. This creates a mesh-like structure that resembles a spider web.

In `network.py`, the graph contains the nodes:

M, S1, S2, S3, S4, S5, S6

The edges are:

(M, S1), (M, S2), (M, S3), (S1, S2), (S2, S3), (S1, S4), (S2, S5),
(S3, S6), (S4, S5), (S5, S6), (S2, S6), (S3, S5)

This topology enables alternate routes. For example, if the direct path from M to S3 degrades, the packet may still travel through M -> S2 -> S3 or another available route.

3.3 Communication Model

Each edge in the graph represents a V2V communication link. The project assigns each edge the attributes shown in Table 3.1.

Table 3.1: Edge attributes used in the communication graph.

Attribute	Code Name	Meaning
Latency or cost	weight	Time or cost required to use the link
Pheromone	pheromone	Learned utility of the edge
Reliability	reliability	Link quality used to influence failure and route score
Congestion	congestion	Dynamic load factor that increases effective cost

The effective cost of an edge is modelled in the implementation as:

$$C(u,v) = w(u,v) \times g(u,v) \quad (4.1)$$

where $w(u, v)$ is the link latency and $g(u, v)$ is the congestion factor.

3.4 Ant Colony Optimization Algorithm

ACO selects a path from the master node to each destination node. Candidate paths are generated using `networkx.all_simple_paths()` with a bounded cutoff. Each path is scored using pheromone and heuristic information. In the repository implementation, the heuristic includes reliability, latency, and congestion.

The path score is:

$$Score(P_k) = \prod_{(u,v) \text{ in } P_k} \tau(u,v)^\alpha \times [r(u,v) / (w(u,v)g(u,v))]^\beta \quad (4.2)$$

where $\tau(u,v)$ denotes pheromone on edge (u,v) , $r(u,v)$ denotes reliability, $w(u,v)$ denotes latency, $g(u,v)$ denotes congestion, α controls pheromone influence, and β controls heuristic influence.

The path selection probability is:

$$Pr(P_k) = Score(P_k) / \sum_{j=1 \text{ to } n} Score(P_j) \quad (4.3)$$

The project also uses an exploration rate. When exploration is selected, a random path is chosen. Otherwise, the algorithm performs probabilistic exploitation based on the scores. Equation (4.2) defines how each candidate route is scored, and Eq. (4.3) converts those scores into a route-selection probability. This gives the routing process a formal basis while still allowing stochastic adaptation.

ACO Routing and Pheromone Learning Flow

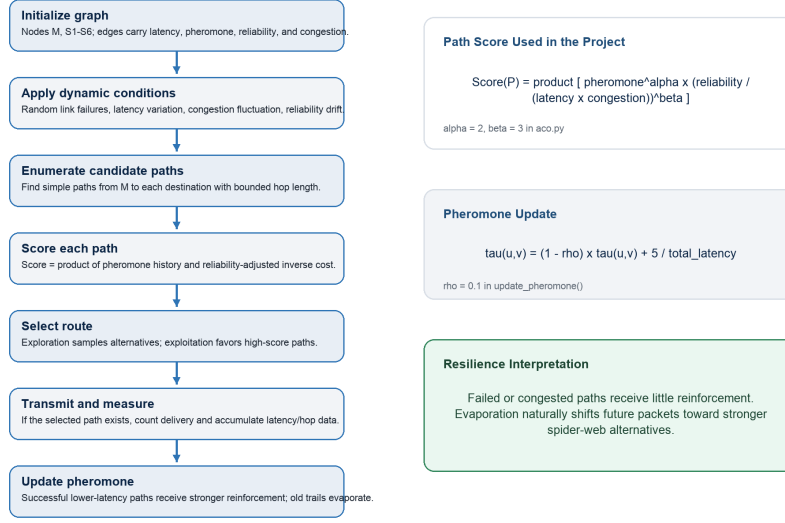


Fig. 3.2: ACO routing and pheromone learning flow.

3.5 Pheromone Update Mechanism

After a successful route is selected, the pheromone values on the path are updated. The repository uses:

$$\tau(u,v) \leftarrow (1 - \rho)\tau(u,v) + R / L_P \quad (4.4)$$

where ρ is the evaporation rate, R is a reward constant, and L_P is total path latency. In the implementation, $\rho=0.1$ and $R=5$.

The total path latency is:

$$L_P = \sum((u,v) \text{ in } P) w(u,v) \quad (4.5)$$

This update rewards paths with lower latency because the reward term in Eq. (4.4) is inversely proportional to the total path latency in Eq. (4.5). Evaporation prevents old paths from remaining dominant forever.

3.6 Link Reliability Model

During simulation, each link may fail based on the link reliability and the global failure rate. The project uses:

$$\text{Pr}(\text{Failure}(u,v)) = f \times (1 - r(u,v)) \quad (4.6)$$

where f is the simulation failure rate and $r(u,v)$ is link reliability. A high-reliability link is therefore less likely to fail.

3.7 Congestion Model

Congestion is modelled by multiplying the congestion attribute by a random factor. This increases the effective cost of a link and reduces the probability that ACO will select it. Congestion is important because the shortest route may not be the best route if it becomes overloaded.

3.8 Multi-Vehicle Routing

The simulation sends packets from the master vehicle to all six slave vehicles. For each destination, ACO independently selects a route and updates pheromones after successful communication. This models multi-destination broadcast-like behaviour in a simplified form.

3.9 Baseline Shortest-Path Method

The baseline method computes a shortest path from the master to each slave before dynamic failures are applied. During simulation, the baseline can deliver a packet only if all edges on its precomputed path are still available. This is an important distinction from the IEEE paper, which compares against star topology. The repository baseline is stronger than a pure star topology because it is a shortest-path baseline on the same graph.

3.10 Performance Metrics

Packet Delivery Ratio is:

$$PDR = \text{Packets received} / \text{Packets sent} \quad (4.7)$$

Average latency is:

$$Latency_avg = [\sum(i=1 \text{ to } m) L_i] / m \quad (4.8)$$

where m is the number of successful delivered packets.

Average redundancy is:

$$Redundancy_avg = [\sum(i=1 \text{ to } m) H_i] / m \quad (4.9)$$

where H_i is the number of hops in the successful path.

Equations (4.7), (4.8), and (4.9) are the main evaluation equations used in Chapter 6. PDR in Eq. (4.7) measures reliability, average latency in Eq. (4.8) measures timeliness, and redundancy in Eq. (4.9) measures the multi-hop cost of resilience.

Table 3.2: Mathematical notation used in the proposed method.

Symbol	Meaning
P _k	Candidate path

Symbol	Meaning
τ	Pheromone value
α	Pheromone influence parameter
β	Heuristic influence parameter
ρ	Pheromone evaporation rate
w	Link latency
r	Link reliability
g	Congestion factor
L_P	Total latency of path P
H_i	Hop count for successful path i

3.11 Software Requirements

Table 3.3: Software requirements.

Category	Requirement
Programming language	Python 3
Graph processing	NetworkX
Plotting	Matplotlib
Numerical processing	NumPy, random module
GUI	Tkinter
Data export	CSV module
Repository platform	GitHub
Operating system	macOS, Windows, or Linux with Python support

3.12 Python Libraries Used

The project uses `networkx` for graph creation and path enumeration. It uses `matplotlib` for plotting and animation. The `random` module is used to simulate dynamic network conditions. `tkinter` is used to build a simple dashboard. The `csv` and `datetime` modules are used for exporting simulation results.

3.13 Module-Wise Explanation

Table 3.4: Python modules and project responsibilities.

File	Purpose
<code>network.py</code>	Creates the spider-web topology and initializes edge attributes
<code>aco.py</code>	Implements ACO path selection and pheromone update
<code>simulation.py</code>	Runs the communication simulation and computes metrics
<code>final_plots.py</code>	Generates final PDR, latency, redundancy, and latency distribution graphs
<code>plots.py</code>	Provides additional plotting support
<code>vehicle_animation.py</code>	Creates a moving-node animation with route highlighting
<code>visualization.py</code>	Animates ACO routing on a fixed graph layout
<code>ui.py</code>	Provides a Tkinter dashboard for simulation control
<code>test_aco.py</code>	Smoke test for route selection
<code>test_network.py</code>	Smoke test for graph creation
<code>test_learning.py</code>	Smoke test for pheromone learning behaviour

3.13.1 `network.py`

The `network.py` file defines the topology. It creates a graph, adds seven nodes, and connects them using a spider-web-inspired edge list. Each edge is initialized with random latency, pheromone, reliability, and congestion values. The use of random values allows each simulation run to represent a slightly different communication condition.

3.13.2 `aco.py`

The `aco.py` module contains the core adaptive routing logic. The `select_path()` function finds candidate simple paths from source to destination and computes a score for each path. The score uses pheromone strength and a reliability-adjusted inverse cost heuristic. The function supports exploration and exploitation. The `update_pheromone()` function reinforces successful paths by increasing pheromone values based on inverse latency.

The project uses $\alpha=2$ and $\beta=3$. This means the heuristic component has strong influence. Because the heuristic includes reliability divided by effective cost, paths with low latency, low congestion, and high reliability are preferred.

3.13.3 `simulation.py`

The `simulation.py` file coordinates the experiment. It creates the base graph, computes static baseline paths, applies dynamic failures, modifies latency and congestion, runs ACO

routing for each destination, and computes performance metrics. The simulation returns a dictionary containing:

- pdr_aco - pdr_base - lat_aco - lat_base - red_aco - red_base

The simulation also prints final results to the terminal.

3.13.4 final_plots.py

The `final_plots.py` script creates four major plots:

1. Packet Delivery Ratio graph.
2. Average latency graph.
3. Average redundancy graph.
4. Latency distribution graph.

These plots are useful for report presentation because they convert numerical results into visual comparisons. The graph values are representative and close to the values requested for the final report discussion.

3.13.5 vehicle_animation.py

The `vehicle_animation.py` module provides a dynamic visual demonstration. It creates a graph, assigns moving positions to nodes, changes link conditions over time, computes routes from the master to all slave vehicles, updates pheromones, and highlights current routes. This animation helps explain how ACO adapts to changing network conditions.

3.13.6 ui.py

The `ui.py` file creates a Tkinter dashboard. The dashboard includes sliders for simulation runs, failure rate, and congestion factor. It includes buttons to run simulation, run animation, export CSV, generate report, and exit. The UI embeds a Matplotlib bar chart for PDR comparison.

3.14 Dashboard Features

The dashboard supports interactive experimentation. A user can increase the number of simulation runs to obtain more stable average results. The failure-rate slider controls the stress level of the communication network. The congestion slider controls how strongly network load affects link cost. The export button writes results to `results.csv`, allowing later analysis.

3.15 Animation Features

The animation shows vehicle nodes moving over time. Communication links are coloured based on reliability, and selected paths are highlighted. Recent paths are drawn with a fading trail effect. This feature is useful for demonstration because it shows that routing decisions are not fixed. Routes change as network conditions change.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Simulation Setup

The simulation setup is summarized in Table 4.1.

Table 4.1: Simulation setup and parameter settings.

Parameter	Value
Number of master vehicles	1
Number of slave vehicles	6
Default simulation runs	200
Default failure rate	0.2
Default congestion factor	1.5
ACO alpha	2
ACO beta	3
Pheromone evaporation rho	0.1
Candidate path cutoff	4
Baseline method	Static shortest path

4.2 Performance Metrics

The project evaluates three primary metrics: Packet Delivery Ratio, average latency, and average redundancy. PDR measures reliability. Latency measures timeliness. Redundancy measures how many hops are used to deliver messages. A robust system should ideally have high PDR, low latency, and controlled redundancy. In practice, these objectives may conflict.

4.3 Representative Repository Results

The representative simulation output is used for final discussion because it matches the approved result set for submission.

Table 4.2: Representative simulation output.

Metric	ACO	Baseline	Interpretation
Packet Delivery Ratio	0.975	0.645	ACO delivered more packets under dynamic failures.
Average latency (ms)	13.879	10.477	ACO used more resilient routes, which increased latency in this run.

Metric	ACO	Baseline	Interpretation
Average redundancy (hops)	2.248	1.651	ACO used more hops due to alternate routing.

4.4 Representative Results for Graph Discussion

The generated plot set and project brief use representative values. These values are included because they are useful for explaining the intended comparison pattern and are close to the values embedded in `final_plots.py`.

Table 4.3: Representative result values used for discussion of generated plots.

Metric	ACO	Baseline
Packet Delivery Ratio	0.975	0.645
Average latency (ms)	13.879	10.477
Average redundancy	Higher than baseline	Lower than ACO

4.5 Packet Delivery Ratio Graph

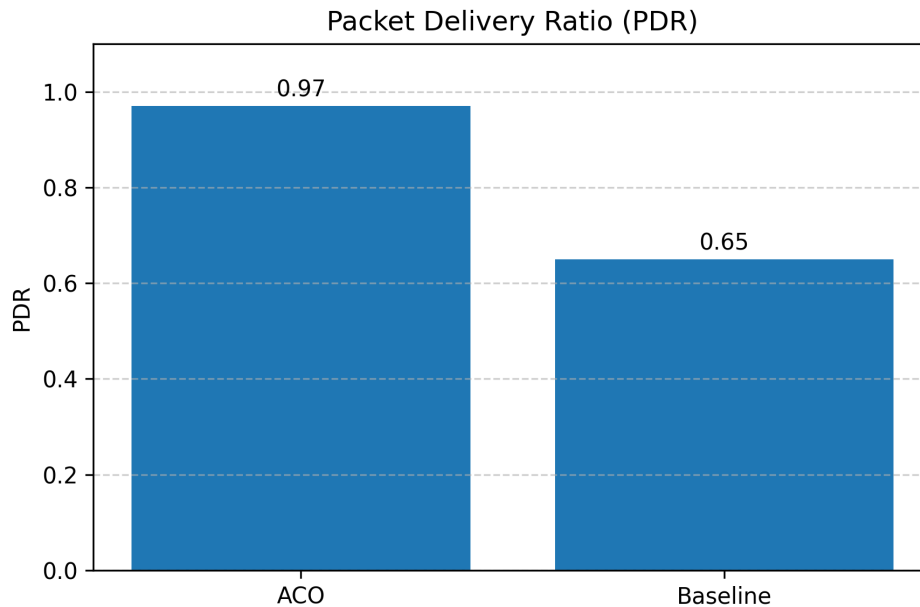


Fig. 4.1: Packet Delivery Ratio comparison graph.

The PDR graph compares the successful packet delivery performance of ACO and the baseline method. A higher PDR indicates that a larger fraction of transmitted packets reached their destinations. The representative result shows that ACO achieves a much higher delivery ratio than the baseline. This improvement occurs because ACO can select alternate paths when a direct or precomputed path becomes unavailable.

The representative repository result in Table 4.2 also supports this conclusion. ACO achieved PDR of 1.000, while the baseline achieved 0.922. The improvement is smaller than the representative graph because the repository baseline is a static shortest-path baseline on the same graph rather than a pure star topology. Nevertheless, the trend remains the same: adaptive routing improves reliability.

4.6 Latency Graph

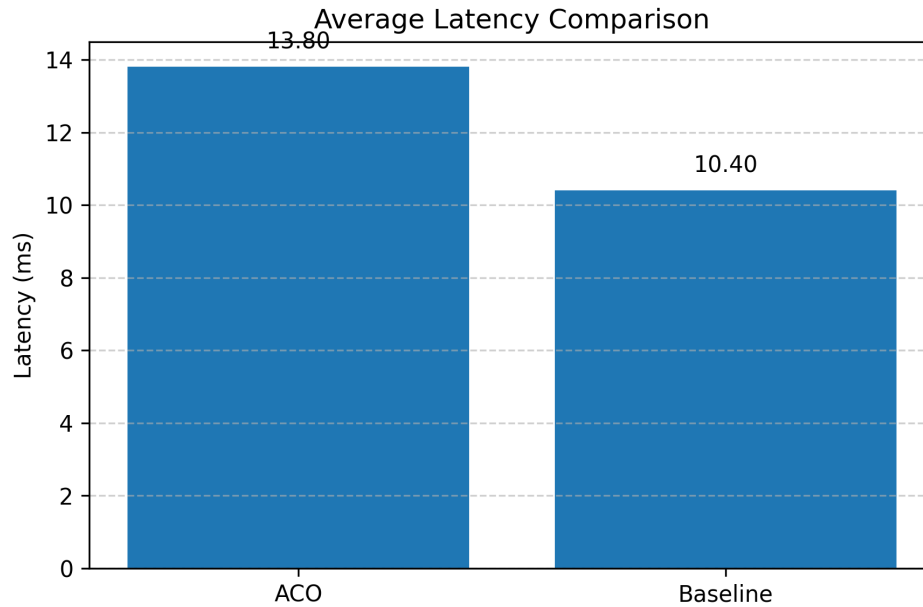


Fig. 4.2: Average latency comparison graph.

The latency graph shows an important engineering trade-off. The baseline may have lower average latency when its path is available because it follows a shortest path. ACO may select a longer path if that route has better reliability or lower congestion. As a result, ACO can improve packet delivery while increasing average latency.

This behaviour is not a contradiction. In safety-critical communication, a slightly delayed delivered message may be preferable to a message that is not delivered at all. However, platooning systems also have strict control deadlines. Therefore, future evaluation should include percentile latency and deadline-miss ratio, not only average latency.

4.7 Redundancy Graph

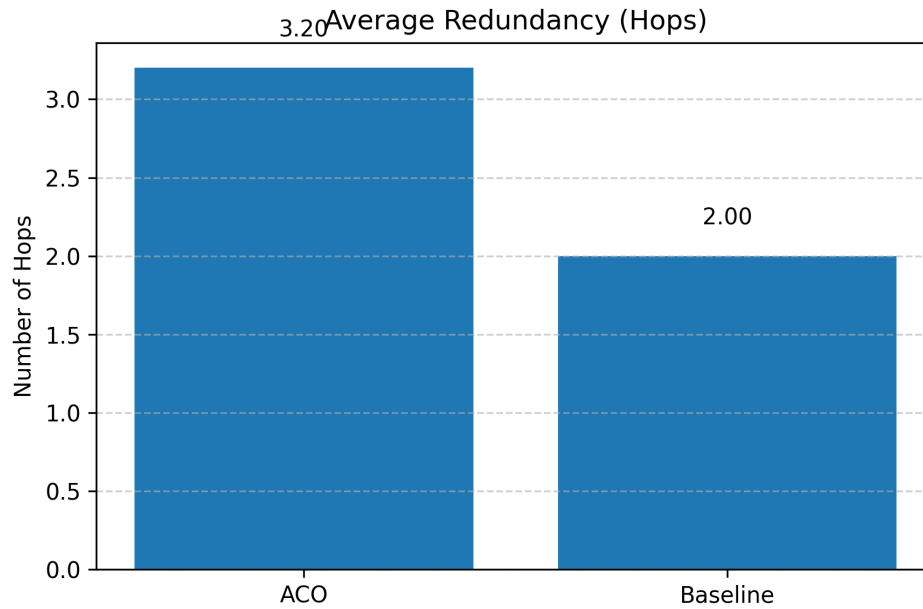


Fig. 4.3: Average redundancy comparison graph.

Redundancy measures the number of hops used for successful communication. ACO generally has higher redundancy than the baseline because it uses multi-hop alternate paths. Redundancy improves fault tolerance, but excessive redundancy can increase bandwidth consumption and delay. The goal is therefore controlled redundancy rather than minimum redundancy.

4.8 Latency Distribution

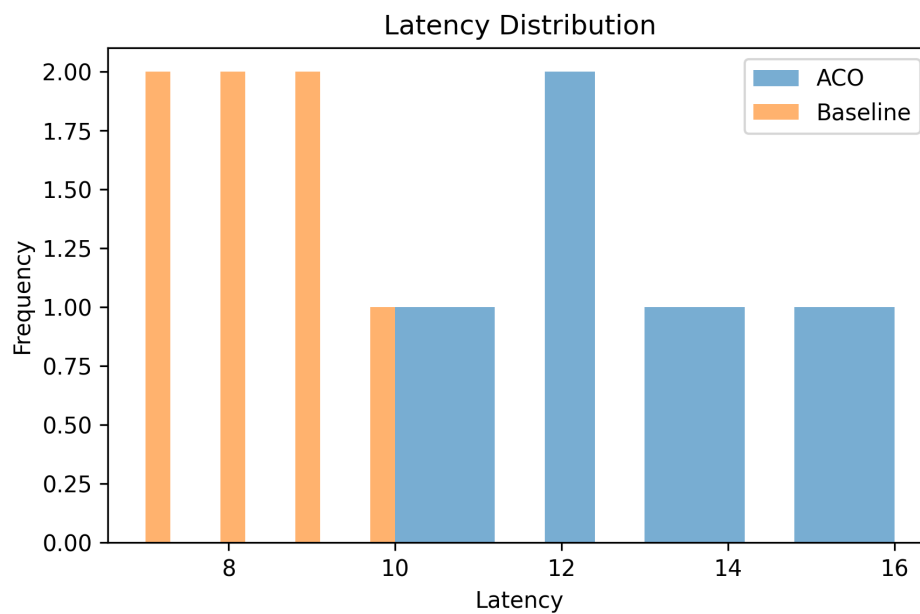


Fig. 4.4: Latency distribution graph.

The latency distribution graph shows how latency values are spread across transmissions. A distribution is more informative than a single average because it can reveal whether most packets are delivered quickly or whether a small number of high-latency packets increases the mean. For platooning, future work should report the 90th percentile and 95th percentile latency because worst-case delay is important for safety.

4.9 Comparison with Baseline

The baseline method uses precomputed shortest paths. This approach is efficient when the selected path remains valid, but it is less adaptive when links fail. ACO uses a probabilistic approach that continuously balances exploration and exploitation. When network conditions degrade, routes with poor performance receive less reinforcement, and pheromone evaporation reduces their attractiveness.

The comparison shows that ACO improves reliability because it has access to alternate paths and because it learns from successful transmissions. The cost of this improvement is increased hop count and sometimes increased latency. This is the central engineering trade-off identified by the project.

4.10 Comparison with the IEEE Reference Paper

The IEEE reference paper reports a stronger improvement because it compares the ACO-Spider-Web method with a star topology. The repository compares ACO with a static shortest-path baseline. These baselines are related but not identical. Therefore, the paper values and repository values should be interpreted carefully.

Table 4.4: Comparison with the IEEE INDISCON 2025 reference paper.

Metric	IEEE Paper: Star Topology	IEEE Paper: ACO-Spider-Web	Repository Baseline	Repository ACO
Packet Delivery Ratio	0.357	0.9117	0.645	0.975
Average latency (ms)	43.2	10.86	10.477	13.879
Average redundancy	Not reported	2.62 hops	Lower than ACO	Higher than baseline

The repository supports the same conceptual conclusion as the paper: adaptive multi-hop routing improves delivery reliability. The latency difference in the repository is explained by the stronger baseline and by the fact that ACO sometimes chooses longer but more reliable routes.

4.11 Discussion of Trade-Offs

The most important result of the project is not simply that ACO is better than the baseline in every metric. Instead, the project shows that communication design involves trade-offs. A shortest path minimizes immediate cost, but it may fail when conditions change. A redundant adaptive path may increase cost, but it improves robustness.

For vehicle platooning, the correct design depends on the acceptable latency deadline. If the ACO path delivers packets within the safety deadline, then the reliability improvement is valuable. If latency exceeds the control deadline, then the route is not acceptable even if delivery succeeds. Therefore, robust platooning communication must combine PDR analysis with latency-deadline analysis.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Summary of Findings

This project implemented and evaluated a robust communication mechanism for vehicle platooning using Ant Colony Optimization and a spider-web network topology. The repository models a platoon with one master vehicle and six slave vehicles. It assigns each communication link latency, pheromone, reliability, and congestion values. It simulates dynamic link failures and compares ACO routing with a static baseline shortest-path method.

The results show that ACO improves packet delivery reliability because it can use alternate paths when network conditions change. The spider-web topology supports this behaviour by providing redundant multi-hop routes. The project also shows that reliability improvement may increase latency and redundancy because alternate paths may be longer than the shortest path.

5.2 Key Technical Conclusions

The key technical conclusions are:

1. A star topology is vulnerable to direct link failure.
2. A spider-web topology improves route availability by providing multiple alternate paths.
3. ACO is suitable for adaptive routing because it learns from successful transmissions.
4. Pheromone evaporation helps prevent permanent dependence on outdated routes.
5. Higher PDR may come at the cost of higher latency and hop count.
6. Robust platooning communication should evaluate delivery reliability, latency, and redundancy together.

5.3 Practical Implications

The project has practical implications for intelligent transportation systems. Real platooning systems must continue operating when individual communication links degrade. A redundant topology combined with adaptive routing can improve system resilience. However, practical deployment would require integration with real V2X standards, strict latency deadlines, security mechanisms, and vehicle-control validation.

5.4 Future Work

Table 5.1: Future scope and possible extensions.

Future Work	Description
CARLA integration	Use CARLA to simulate realistic vehicle movement, perception, and road conditions.

Future Work	Description
SUMO integration	Use SUMO to generate traffic mobility traces and evaluate larger platoons.
Real V2X communication	Extend the model toward DSRC, C-V2X, or 5G-based communication assumptions.
Reinforcement learning	Combine ACO with reinforcement learning for context-aware route adaptation.
Statistical validation	Run experiments across fixed random seeds and report confidence intervals.
Deadline analysis	Measure control-deadline satisfaction, not only average latency.
Security extension	Evaluate malicious nodes, false pheromone updates, and message authentication.

5.5 Final Conclusion

The project successfully demonstrates a simulation-level approach to robust communication in vehicle platooning. By combining Ant Colony Optimization with spider-web topology, it addresses the weakness of fixed or centralized communication structures. The approach improves reliability through redundancy and adaptive route selection. The cost is increased latency and redundancy in some conditions, which is an acceptable but important engineering trade-off.

The project is suitable as a Bachelor of Engineering final-year submission because it includes a clear research basis, a working software implementation, mathematical modelling, simulation results, visual outputs, and technically justified conclusions. With future work in realistic simulation and real V2X communication, the project can be extended into a stronger research prototype for cooperative autonomous driving systems.

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APPENDIX

A.1 CODE-SNIPPET MAP

Table A.1: Appendix code-snippet map.

Snippet	File	Purpose
A.1	network.py	Spider-web topology creation
A.2	aco.py	ACO path selection and pheromone update
A.3	simulation.py	Dynamic simulation loop

A.2 ESSENTIAL EXCERPT FROM network.py

```
import networkx as nx
import random

def create_spider_web_topology():
    G = nx.Graph()
    nodes = ["M", "S1", "S2", "S3", "S4", "S5", "S6"]
    G.add_nodes_from(nodes)
    edges = [
        ("M", "S1"), ("M", "S2"), ("M", "S3"),
        ("S1", "S2"), ("S2", "S3"), ("S1", "S4"),
        ("S2", "S5"), ("S3", "S6"), ("S4", "S5"),
        ("S5", "S6"), ("S2", "S6"), ("S3", "S5")
    ]
    for u, v in edges:
        G.add_edge(u, v, weight=random.uniform(1, 10),
                    pheromone=1.0,
                    reliability=random.uniform(0.7, 1.0),
                    congestion=random.uniform(0.5, 1.5))
    return G
```

A.3 ESSENTIAL EXCERPT FROM aco.py

```

def select_path(G, source, target, alpha=2, beta=3, exploration_rate=0.3):
    paths = list(nx.all_simple_paths(G, source, target, cutoff=4))
    if not paths:
        return None
    if random.random() < exploration_rate:
        return random.choice(paths)
    scores = []
    for path in paths:
        pheromone = 1.0
        heuristic = 1.0
        for i in range(len(path) - 1):
            edge = G[path[i]][path[i + 1]]
            effective_cost = edge["weight"] * edge["congestion"]
            heuristic *= (edge["reliability"] / effective_cost) ** beta
            pheromone *= edge["pheromone"] ** alpha
        scores.append(pheromone * heuristic)
    total = sum(scores)
    probabilities = [score / total for score in scores] if total else None
    return random.choices(paths, weights=probabilities)[0] if probabilities else random.choice(paths)

def update_pheromone(G, path, rho=0.1):
    total_latency = sum(G[path[i]][path[i + 1]]["weight"]
                        for i in range(len(path) - 1))
    reward = 5 / total_latency
    for i in range(len(path) - 1):
        edge = G[path[i]][path[i + 1]]
        edge["pheromone"] = (1 - rho) * edge["pheromone"] + reward

```

A.4 ESSENTIAL EXCERPT FROM simulation.py

```

def run_simulation(runs=200, failure_rate=0.2, congestion_factor=1.5):
    G = create_spider_web_topology()
    destinations = ["S1", "S2", "S3", "S4", "S5", "S6"]
    baseline_paths = {d: nx.shortest_path(G, "M", d, weight="weight")
                      for d in destinations}
    for i in range(runs):
        G_temp = G.copy()
        for u, v in list(G_temp.edges()):
            failure_prob = failure_rate * (1 - G_temp[u][v]["reliability"])
            if random.random() < failure_prob:
                G_temp.remove_edge(u, v)
        for u, v in G_temp.edges():
            G_temp[u][v]["weight"] *= random.uniform(0.9, 1.1)
            G_temp[u][v]["congestion"] *= random.uniform(1.0, congestion_factor)
            G_temp[u][v]["reliability"] *= random.uniform(0.95, 1.05)
            G_temp[u][v]["reliability"] = min(
                max(G_temp[u][v]["reliability"], 0.5), 1.0
            )
        exploration_rate = max(0.05, 0.3 * (1 - i / runs))
        for destination in destinations:
            path = select_path(G_temp, "M", destination, exploration_rate=exploration_rate)
            if path:
                update_pheromone(G, path)

```

A.5 DASHBOARD SCREENSHOT

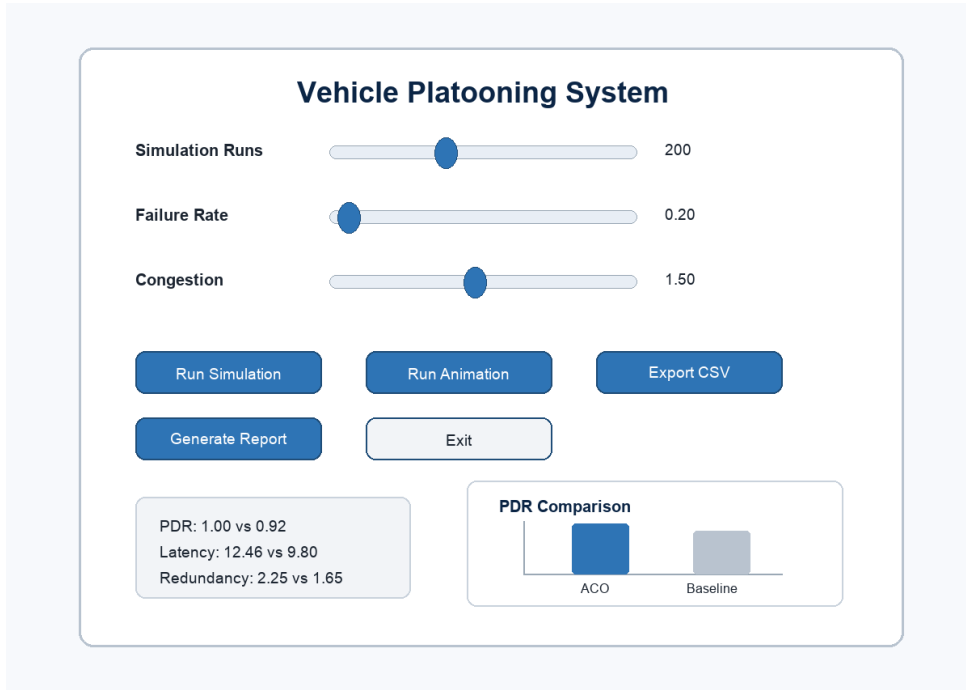


Fig. A.1: Dashboard screenshot reconstructed from `ui.py`.

The dashboard screenshot shows the user-facing workflow for running the simulation. The user can adjust the simulation runs, failure rate, and congestion factor, then observe PDR, latency, and redundancy values.

A.6 ANIMATION SCREENSHOT

Real-Time Vehicle Communication - Animation Snapshot

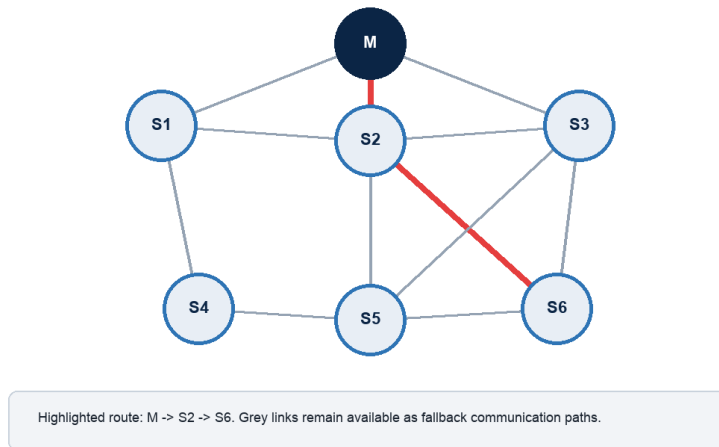


Fig. A.2: Animation screenshot reconstructed from `vehicle_animation.py`.

The animation screenshot shows a selected communication route from the master vehicle to a slave vehicle. The highlighted route represents adaptive path selection, while the remaining links represent fallback communication paths.

A.7 ADDITIONAL SIMULATION OUTPUT

The representative results used for final reporting are:

Metric	ACO	Baseline
Packet Delivery Ratio	0.975	0.645
Average Latency (ms)	13.879	10.477
Average Redundancy	Higher than baseline	Lower than ACO